



Hands-on Activities for Digital Forensics Education

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ABSTRACT

As teaching professors we have noticed that hands-on activities for a digital forensics course fall into the following three categories: 1) activities that assist students in learning how to use common digital forensics tools; 2) activities that help students gain in-depth understanding of the basic concepts and fundamental knowledge that are presented in class lectures; 3) activities that promote students the development of mindsets and data analytical skills that are needed for a digital forensic investigator. Various formats are employed to develop these hands-on exercises in different categories. The educational objectives and student learning outcomes map well to the CAE-CD (Centers of Academic Excellence - Cyber Defense) outcomes by completing their forensic knowledge units.

In this paper, we share our idea and experience to design and implement such hands-on assignments in each category for meeting specific educational objectives. Sample exercises are briefly described to explain our idea in each category. Open-source tools and data sets are introduced for reference. Experiences, lessons, and sample feedback from students are discussed. Our results will provide a point of reference for those who teach digital forensics courses at a college or university or are developing a digital forensic curriculum.

CCS CONCEPTS

• **Security and privacy** → **Systems security**; *Network security*; Database and storage security; Privacy protections.

KEYWORDS

Digital Forensics; Education; Lab; Hands-on Activities

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1 INTRODUCTION

Through an extensive review of existing literature in digital forensics education, we have discovered the lack of clear answers to the following questions:

- (1) What kinds of hands-on activities are needed for a course on digital forensics?
- (2) What are the educational objectives?
- (3) What are the specific learning outcomes associated with each lab activity?

With many years of experience of teaching digital forensics at both undergraduate and graduate levels in a 4-year college, we have developed hands-on lab assignments in three different categories as briefly described below:

- (1) **Tool Use:** Students will learn how to use the tools that are used to complete the tasks for a digital forensic investigation through hands-on lab activities in this category.
- (2) **Knowledge Reinforcement:** Labs in this category are designed to reinforce the basic concepts and fundamental knowledge that are presented in class lectures.
- (3) **Mindset Development:** Lab assignments in this category are designed to help students develop the mindset that is necessary to be an effective digital forensics investigator.

Although we update and develop new labs every year, all updated and newly developed labs (over several years) fall into these three categories.

In this paper, we will explain the ideas to design and develop such lab assignments for a digital forensics course, describe the educational objectives of the lab assignments in each category, elucidate how the learning outcomes of our lab assignments can be mapped to the outcomes through the completion of forensics knowledge units adopted by the CAE-CD (Centers of Academic Excellence - Cyber Defense) knowledge units [2]. Results of this work will be helpful to those who are teaching or going to teach a digital forensics course at a college or a university. Our experience can also be a point of reference for the curriculum development in digital forensics.

2 DESIGN AND DEVELOPMENT OF HANDS-ON ACTIVITIES

In this section, we will discuss the design and development of hands-on activities in the three categories that were briefly introduced previously.

2.1 Labs for Tool Use

We have introduced a variety of software tools in the domain of free software that are widely used by digital forensic investigators and IT professionals. Examples include **FTK® Imager** for the creation of forensic friendly image of a physical disk, **Autopsy®** for general forensics examination and investigation, **Volatility®** for memory forensics investigation, Access Data **Registry Viewer** for

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Windows registry forensics, and **OpenPuff** as a steganography tool to investigate photographic files that may have hidden data.

The main educational objectives of the labs in this category are to help student learn how to use common forensic tools to examine a dataset and understand the functions a tool provides. In a lab for these purposes, we usually start with a high-level introduction to the tool. Then, a web link to the user manual (if available) is given. Then, we will give the download web page and ask students to download it directly from the Web and install it themselves. After software installation, detailed instructions are given with a dataset to help students get started with this tool. Screenshots with highlighted marks are used to show students what can be found and from where they can be seen.

2.2 Labs for Knowledge Reinforcement

The main educational objective of the labs in this category is to help students gain in-depth understanding of essential concepts and fundamental knowledge presented in class lectures. Labs are usually given after covering these materials in class. One way to develop such labs is to ask informative results after related information has been observed. For example, when students have observed the sizes of a sector and a cluster along with the physical and logical sizes of a file, a lab question could ask students to calculate the sizes in bytes of drive slack, RAM slack, and file slack.

2.3 Labs for Mindset Development

After the hands-on exercises in the first two categories, we design and develop hands-on activities in this category to help the future digital forensic analysts develop the mindsets they need to perform their work effectively. These activities are designed either as case studies or course projects. Both are implemented as a teamwork. Case studies usually take one or two weeks, while a course project needs 4-6 weeks to complete. A lab of case study consists of a brief description about the case and the requirements to complete the investigation. An existing forensic image is provided, and hints may be given based on the difficulty level of the case.

3 A MAP TO THE CAE-CD OUTCOMES

The knowledge units and their outcomes described in the document of the National Centers of Academic Excellence in Cybersecurity (NCAE-C) [1] are widely accepted as a good reference to design, develop, and deliver a curriculum in cybersecurity [3].

The map of the learning outcomes through these hands-on activities to the CAE-CD outcomes is summarized below:

Labs for tool use:

- “Use one or more common digital forensics tools, such as EnCase, FTK, ProDiscover, Xways, SleuthKit.”
- “Students will be able to understand how to acquire a forensically sound image.”

Labs for knowledge reinforcement:

- “Describe what can/cannot be retrieved from various Operating Systems.”
- “Describe the methodologies used in host forensics.”

Labs for mindset development:

- “Students will be able to understand how to identify forensic artifacts left by attacks.”
- “Describe the steps in performing digital forensics from the initial recognition of an incident through the steps of evidence gathering, preservation and analysis, through the completion of legal proceedings.”

4 CONCLUSION AND DISCUSSION

We have proposed to design and develop hands-on activities in three categories for digital forensics education to meet the needs of digital forensic investigators. These categories include labs for tool use, knowledge reinforcement, and mindset development. The development of these exercises has been briefly described in each category. The educational objectives and learning outcomes of these hands-on activities map well to the CAE-CD outcomes.

5 ACKS

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REFERENCES

- [1] National Centers of Academic Excellence in Cybersecurity (NCAE-C). 2021. 2020 CAE Cyber Defense (CAE-CD) Knowledge Units. CAE Documents Library, online. <https://public.cyber.mil/ncae-c/documents-library/>. Last Accessed in June 2023.
- [2] National Centers of Academic Excellence in Cybersecurity (NCAE-C). 2023. WELCOME TO THE NATIONAL CENTERS OF ACADEMIC EXCELLENCE IN CYBERSECURITY (NCAE-C). Web page, online. <https://public.cyber.mil/ncae-c/>. Last Accessed in June 2023.
- [3] Weifeng Xu, Lin Deng, and Dianxiang Xu. 2022. Towards Designing Shared Digital Forensics Instructional Materials. In *2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, Vol. 1. IEEE, New York, NY, USA, 117–122. <https://doi.org/10.1109/COMPSAC54236.2022.00025>